Progress Brief — Smart Retail → AI-Powered Business Finance & Inventory Analysis System

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Status: work in progress (paused on request). Nothing in your project folder has been changed yet — all new code so far is in Claude's workspace and will be copied into your folder once the app is complete and tested.

1. Inspection of the existing project

Folder	Finding	Decision
Smart-Retail-Analytics-Fixed	Latest version (Apr 16). Flask + MySQL + Bootstrap + Chart.js. 14 templates, 7 tables.	**Chosen as the base project**
Ecommerce db (Top sell when add)/ecommerce_project	Byte-for-byte identical to the base (same app.py, database.sql, templates).	Duplicate — to be moved to _to_delete/
ecommerce_project/ecommerce_project	Older version (Apr 5): older database.sql and admin_orders.html; has a .git folder.	Outdated duplicate — to be moved to _to_delete/

Existing base: customer storefront (home, products, cart, checkout, reviews, orders) plus an admin area (dashboard with 4 KPIs and 2 charts, products, orders, staff & salaries, customer analytics). Only 10 products and 10 sample orders — too little data for finance analytics or forecasting.

2. Transformation plan (agreed architecture)

• Keep the existing stack: **Flask + MySQL + Bootstrap + Chart.js**; add **pandas, NumPy, scikit-learn** for analytics/AI.

• Reuse existing tables (products, orders, order_items, customers, staff, salaries); extend products with SKU, cost price, minimum stock, supplier; add an **expenses** table.

• Replace the shopping storefront with a sidebar-based business dashboard: Dashboard, Revenue & Profit, Cash Flow, Expenses, Inventory, Stock Alerts, Sales Analytics, Demand Forecast, Health Score, AI Insights, Staff & Payroll.

• Staff login (admin@shop.com / admin123) preserved; customer cart/checkout pages retired.

3. Completed so far

File	What it does
seed.py (new)	Deterministic demo-data generator. 12 months (Oct 2025 – Sep 2026): 24 products (original 10 kept + 14 new, e.g. Wireless Mouse), 20 customers incl. B2B, 6 staff with monthly payroll, **4,828 sales orders / 9,448 order lines**, **253 expenses** in 8 categories (Rent, Salaries, Utilities, Marketing, Inventory Purchase, Transportation, Operations, Miscellaneous). Includes festive seasonality, growing/declining products, deliberate low/out-of-stock and overstock items, and planted unusual expenses (Sep marketing spike, Jul AC repair, Feb GST penalty, May electricity spike) for the anomaly detector.
database.sql (regenerated)	Full MySQL 8 schema + all demo data for database **business_finance_db** (importable via phpMyAdmin or mysql CLI).
db.py (new)	Data-access layer. Uses MySQL; auto-creates and seeds the database if empty; falls back to a local SQLite demo DB if MySQL isn't running, so the project can always be demonstrated.

Tested: data generation and loading run successfully. Sample monthly results from the generated data:

Month	Revenue	COGS	Operating exp.	Net profit	Net cash flow
Oct 2025	■17.77 L	■11.40 L	■4.05 L	■2.32 L	■2.29 L
Jun 2026	■16.08 L	■10.21 L	■3.86 L	■2.01 L	■3.64 L
Aug 2026	■15.22 L	■9.58 L	■3.83 L	■1.81 L	■1.50 L
Sep 2026	■16.17 L	■10.19 L	■4.46 L	■1.52 L	−■1.26 L (pre-festive stock-up)

4. Remaining work

Priority	Item
P0	analytics.py engine (revenue, COGS, gross/net profit, margins, growth); new app.py routes; Dashboard (8 KPIs + 7 charts); Expenses (list/add/filter/trends); Revenue & Profit page; Inventory table with stock status; Low-stock alerts; Sales analytics
P1	Demand forecasting (scikit-learn trend × seasonal index, hold-out accuracy, recommended stock, shortage/overstock flags); Financial Health Score (5 factors × 20 pts); rule-based AI recommendations; Cash-flow page with projection; expense anomaly detection (robust Z-score + month-over-month spikes)
P2	New professional UI theme (sidebar layout), CSV exports, filters, README update
Finish	Run and test every page on MySQL and SQLite, fix errors, copy the final project into your folder and move the two duplicate folders to _to_delete/